

AI DIGITAL OFFICE OS

Technical Integrity Review & Required Closure Patch

Baseline: v1.3 Consolidated Master

Review date: 2026-08-30

Executive Verdict

The v1.3 PDF is substantially stronger than v1.2 and preserves the correct overall architecture. It is not yet implementation-lock perfect because several cross-layer contracts are still incomplete. These are structural closure issues, not a reason to redesign the Office.

The four most important blockers are: Project Registry, true Agent Versioning, Agent/Model Run lineage, and complete tenant RLS. Additional high-priority closures are the MCP tool registry, evaluation history, cost ledger, deployment/release records and complete API lifecycle coverage.

Important: v1.3 itself correctly distinguishes a specification from a running production system. Its NFR/SLO values are targets until measured in an actual environment.

Detailed Findings

CRITICAL: Project Registry is missing

Finding: The blueprint and API use `project_id` throughout, but the SQL schema has no `project_registry/projects` table. This breaks referential integrity and prevents the Project Factory from having an authoritative project record.

Required closure: Add `project_registry` with `tenant_id`, `project_id`, `name`, `type`, `constitution_version`, `lifecycle_state`, `risk_level`, `owner`, `repository_ref`, `environment policy`, `created_at` and `updated_at`. Add project endpoints to the API.

CRITICAL: Agent versioning is structurally incomplete

Finding: `agent_registry` uses `agent_id` as the sole primary key while the acceptance criteria require ACTIVE agent updates to create a new version rather than overwrite the active record. A separate `agent_versions` table is required.

Required closure: Add `agent_versions` keyed by `agent_version_id`, with `agent_id`, `version`, `specification_hash`, `prompt_version`, `model_policy`, `permissions snapshot`, `evaluation result`, `status` and `lineage`. Make activation point to a specific version.

CRITICAL: Artifact lineage references undefined run entities

Finding: `artifact_registry` contains `agent_run_id` and `model_run_id`, but there are no `agent_runs` or `model_runs` tables. Full lineage therefore cannot be reconstructed from the relational model.

Required closure: Add `agent_runs` and `model_runs`, plus foreign keys from artifacts, routing/audit records and task attempts.

CRITICAL: RLS is only partially implemented

Finding: The SQL explicitly says to extend RLS later and enables it on only four tables. That is not sufficient for a strong tenant-isolation baseline because prompts, policies, approvals, workflows, messages, memory, secrets, feature flags and RBAC junctions remain exposed to service-layer mistakes.

Required closure: Enable RLS on every tenant-scoped operational table and define both USING and WITH CHECK policies. Keep system-template NULL tenant rows explicitly constrained to template-only tables.

HIGH: Message idempotency is not stored

Finding: The specification requires idempotency for agent messages, but `agent_messages` has no `idempotency_key` column or uniqueness rule.

Required closure: Add `idempotency_key` and `UNIQUE(tenant_id, idempotency_key)`. Store delivery attempts and duplicate-handling outcome.

HIGH: Model evaluation history is missing

Finding: `model_registry` has only a current `evaluation_score`. The blueprint promises versioned, rerunnable model evaluation, but there is no evaluation-run history.

Required closure: Add model_evaluation_runs and model_evaluation_metrics. Preserve benchmark version, model version, evaluator version, score and timestamp.

HIGH: Cost/usage ledger is missing

Finding: Cost governance is defined, but there is no authoritative usage ledger for tokens, requests, provider/model cost or budget consumption.

Required closure: Add usage_events/cost_ledger with tenant, project, task, agent, provider, model, input/output usage, estimated/actual cost, currency, timestamp and billing status.

HIGH: Deployment/release records are missing

Finding: The blueprint has deployment, release, rollback and artifact lineage concepts, but deployment_ref is only a free-text field.

Required closure: Add deployment_registry and release_registry with immutable release identifiers, artifact references, environment, strategy, health state, approval, rollback target and timestamps.

HIGH: MCP tool registry is missing

Finding: MCP is the explicit interoperability standard, but there is no persistent registry for MCP servers/tools, capabilities, allowed actions or tool versions.

Required closure: Add mcp_server_registry, tool_registry and agent_tool_bindings. Bind permissions and policy decisions to stable tool IDs.

HIGH: OpenAPI is still not a complete control-plane contract

Finding: The document says the API is full, but key lifecycle operations are absent or incomplete: project creation, workflow creation/cancel, artifact creation, agent activation/version selection, message send, provider/model registry management and cost/usage queries.

Required closure: Expand /v1 with the missing resource lifecycle endpoints and explicit request/response/error schemas. Protect every endpoint with bearer auth and required scopes.

HIGH: OpenAPI schemas do not fully match SQL

Finding: AgentRecord omits several operational fields present in SQL. Artifact and feature-flag schemas are also narrower than the database model.

Required closure: Generate API schemas from a canonical domain model or maintain a schema-diff gate in CI so SQL and OpenAPI cannot drift.

MEDIUM: Authentication contract needs stronger implementation detail

Finding: JWT bearer authentication is specified, but issuer, audience, JWKS rotation, token expiry, required scopes and service-vs-human claims are not fully defined.

Required closure: Add issuer/audience/JWKS requirements, required scopes, clock-skew policy, token lifetime, service identity claims and key-rotation behavior.

MEDIUM: Session management is mentioned but not modeled

Finding: Layer 9 says session management exists, but there is no session/refresh-token or session-revocation model.

Required closure: Either explicitly delegate sessions to the external identity provider and document that boundary, or add session_registry and revocation semantics.

MEDIUM: Policy decision history deserves a first-class record

Finding: Audit payload can hold policy results, but routing and policy behavior are important enough for deterministic incident reconstruction to deserve structured records.

Required closure: Add policy_decision_records and routing_decision_records linked to task/run IDs, with candidate alternatives and rejection reasons.

MEDIUM: Acceptance claims must distinguish specification from measured evidence

Finding: The PDF correctly says the numeric SLOs are starting baselines, but the checklist sometimes reads like an achieved production state.

Required closure: Label every acceptance item as TARGET, TEST REQUIRED or VERIFIED. No item may be marked VERIFIED until an actual test run produces evidence.

MEDIUM: The master PDF is a consolidated specification, not implementation

Finding: The document honestly states that no infrastructure-as-code, provider contracts, penetration-test results or credentials are included. This is correct, but the next implementation package needs explicit environment contracts.

Required closure: Keep v1.4 as the specification baseline and create a separate implementation phase package containing IaC, deployment topology, secrets integration, real provider adapters and executed test evidence.

What Must NOT Change

Keep the approved hierarchy, Agent Factory, capability-based model routing, multi-provider gateway, Policy Engine, GREEN/YELLOW/RED governance, durable workflows, tiered memory, artifact immutability, independent QA, bounded self-healing, MCP/A2A direction and Project Factory concept.

V1.4 is an integrity patch, not a second architecture.

Implementation-Lock Gate

Lock the blueprint only after the SQL schema, API contract and canonical domain model pass automated consistency checks; all tenant-scoped tables have RLS with USING and WITH CHECK; artifact lineage resolves end-to-end; agent versions are immutable and activatable by version; and acceptance evidence distinguishes TARGET from VERIFIED.

Final decision: V1.3 = architecturally strong. V1.4 integrity closure = required before implementation baseline is frozen.